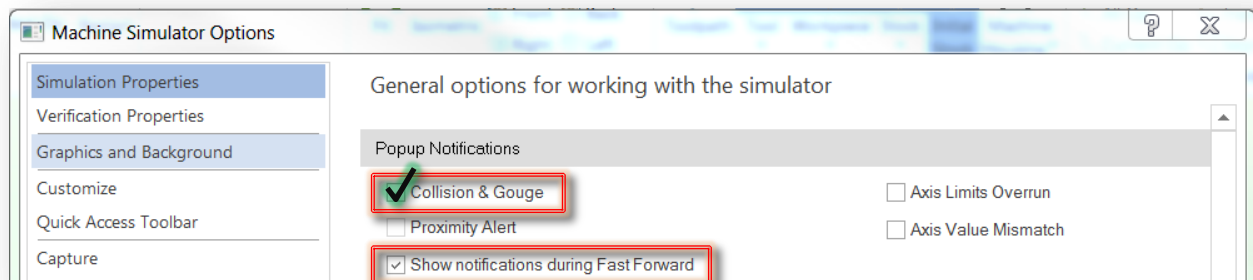
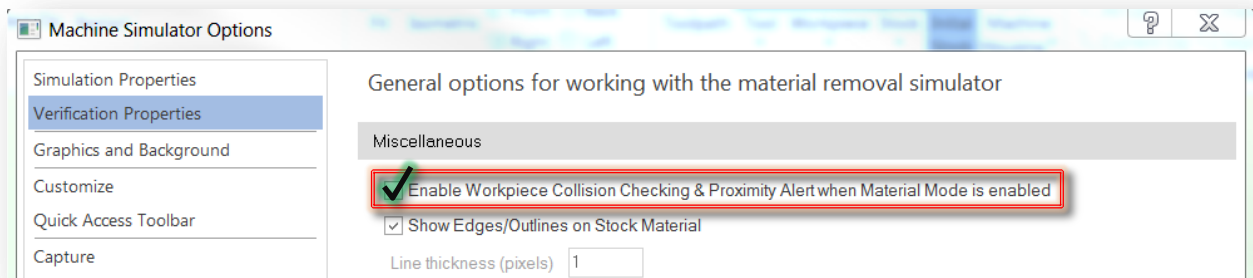
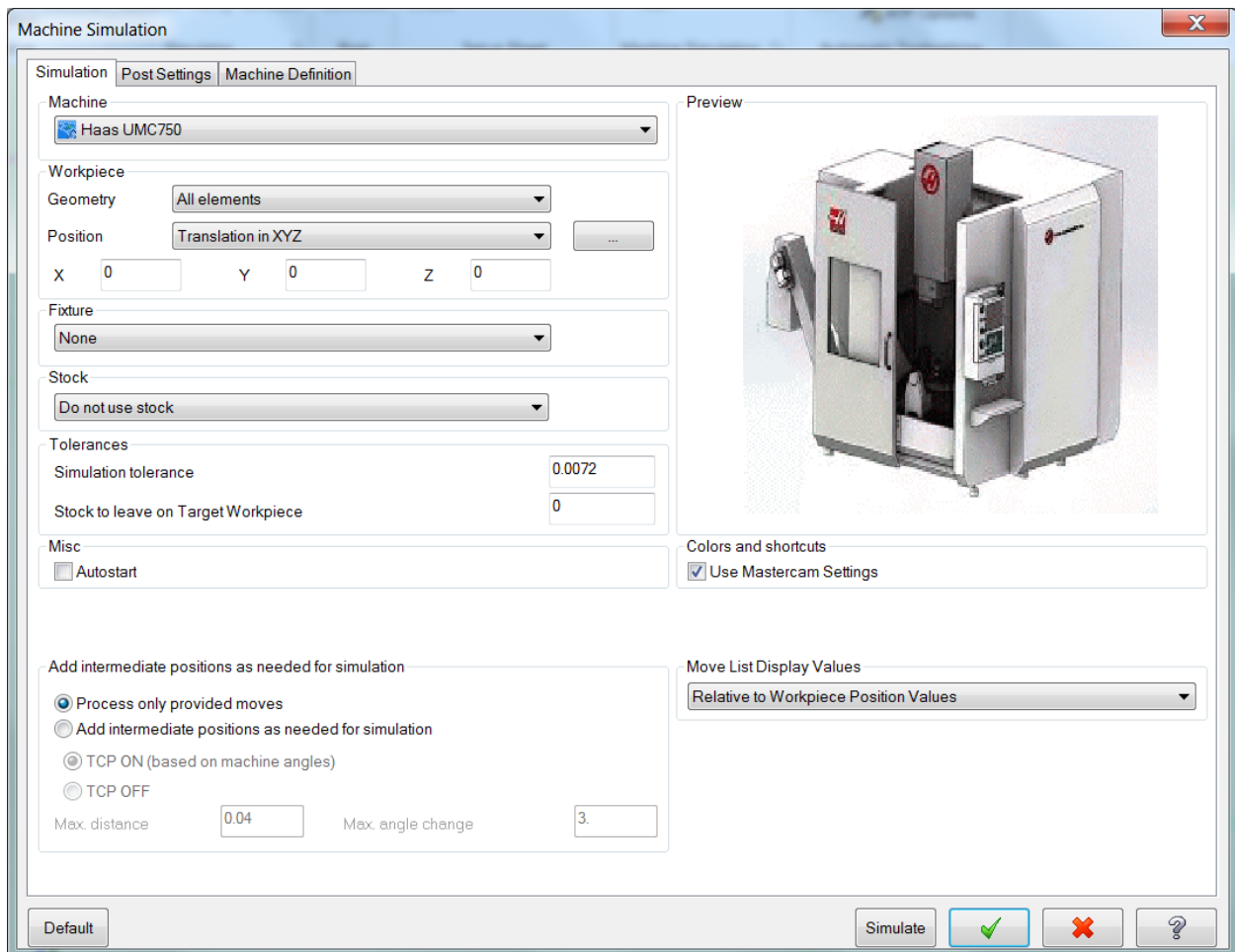
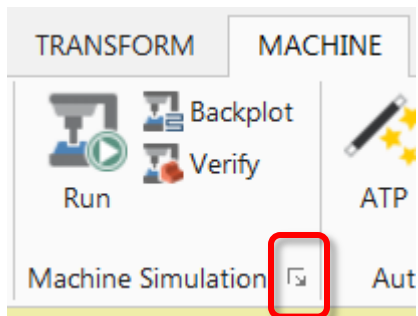


Machine Sim Settings



NOTE: THE PORTIONS OF THE MACHINE THAT ARE CHECKED FOR COLLISION WITH OTHER PORTIONS IS ALSO DETERMINED WITHIN THE MACHINE ITSELF. SEE [THIS SECTION](#) BELOW FOR MORE INFORMATION ON HOW TO CHECK CERTAIN ELEMENTS OF THE MACHINE FOR COLLISION.

Setting Up and Running Machine Sim



Translation in XYZ
Automatic
Translation in XYZ

	MRZP	WORK OFFSET	SHIFT FOR MACHINE SIM	CAD: WCS LOCATION TO MOVE MRZP
X	-11.005	-10.6712	0.334	-0.334
Y	-10.362	-11.1269	-0.7654	0.7654
Z	-19.38	-14.5394	4.8408	-4.8408

MRZP taken directly from the machine setting.

Work offset is values from the work offset page directly

Enter this shift into the dialogue shown below (setup the dialogue to translation in XYZ)

I have had times where I had to switch from "automatic" to "translation XYZ" with a shift of zero in each axis as the automaitc method wasn't positioning it correctly.

Workpiece

Geometry From Level 1

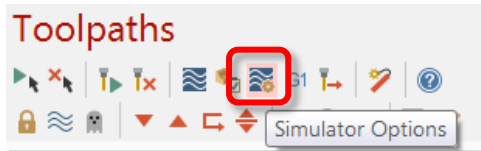
Position Translation in XYZ

X .334 Y -.7654 Z 4.8408

All elements
All elements
Selected elements
From Level
None
Load From STL

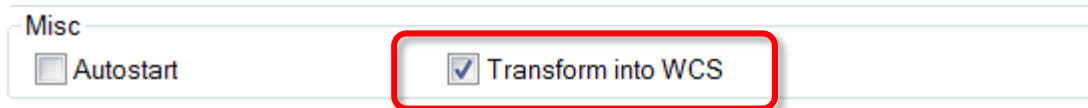
None
Selected elements
From Level
None
Load From STL
Use Mastercam fixture definition

It seems the fixture definition it is referring to is the one set in the current verify settings...

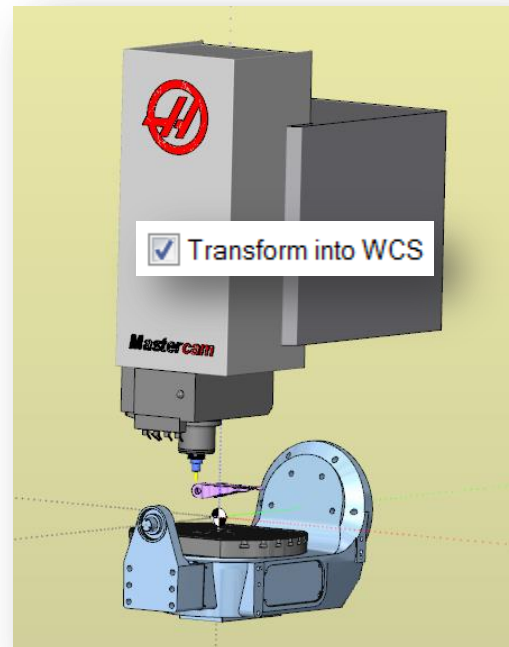
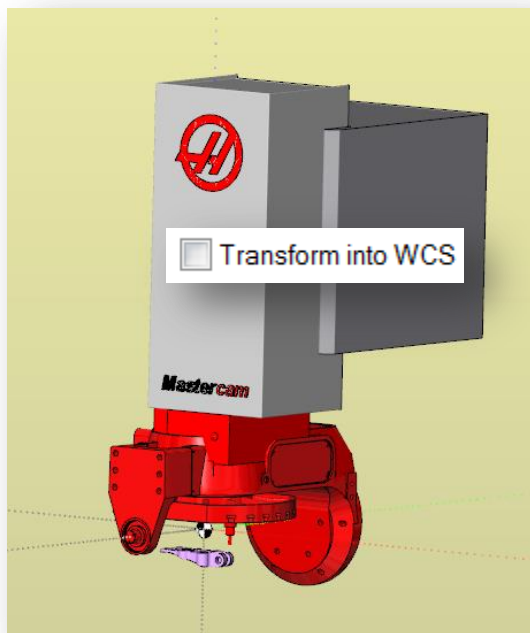


Also note that at one time the fixture did not show, it needed to be added to the machine tree within the machine sim. (See [this section](#) for more info)

If your WCS doesn't correspond to the Global coordinate system "Top" plane (as this is normally the rotary zero degree plane) you will need to check this box.

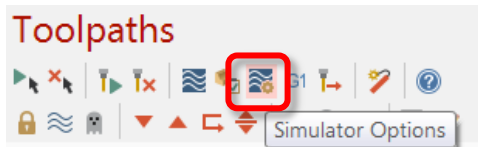
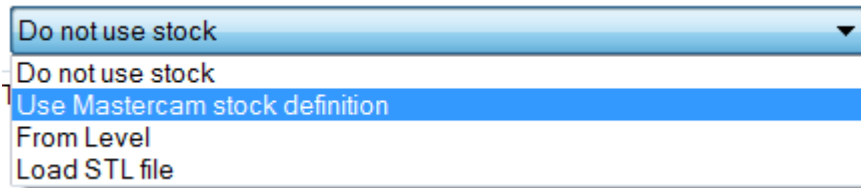


For example: If the trunion starts off at a strange angle this is most likely due to the fact that your WCS is not the same as "World Coordinate"



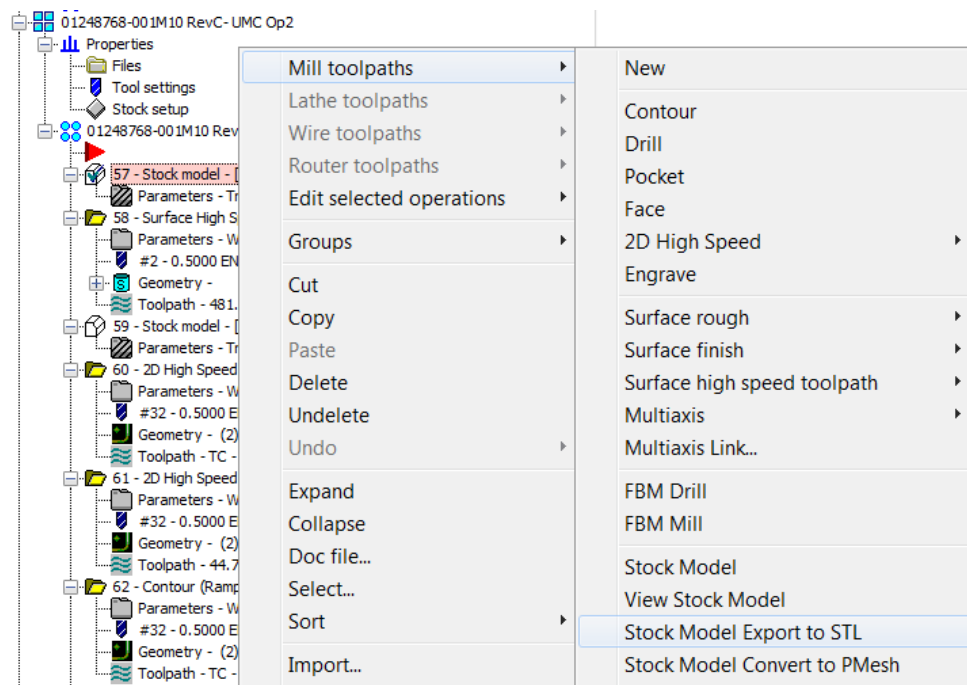
Stock

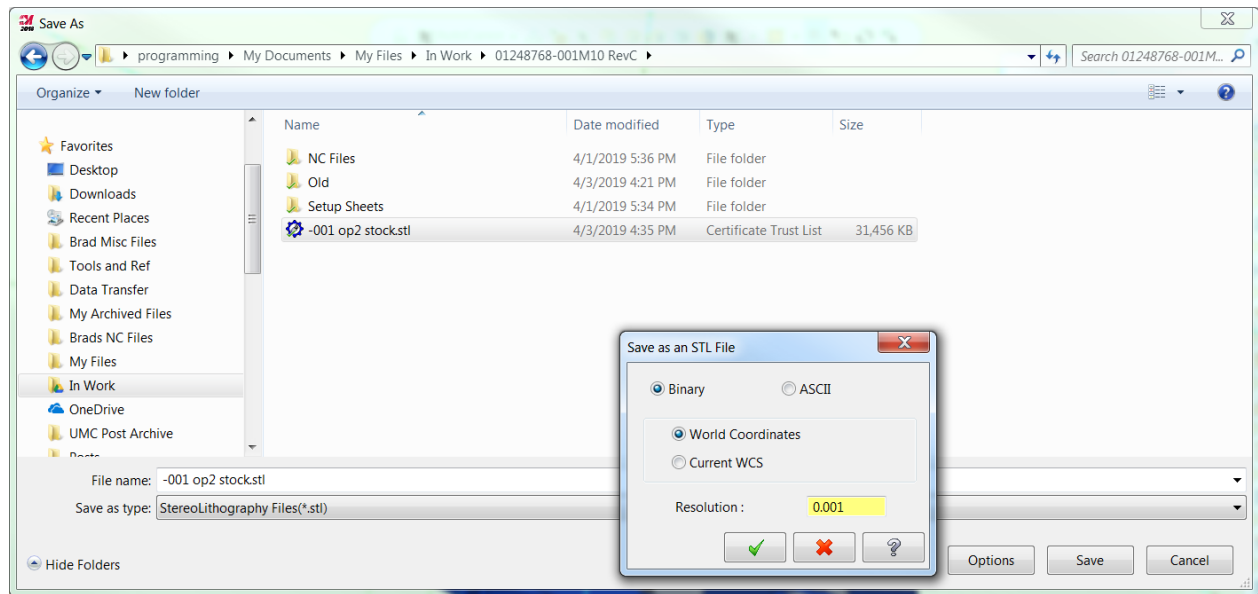
The "mastercam stock definition" seems to be the one found in the Machine group stock properties (as opposed to the one found in the simulation setting found at the top of the toolpath manager as it is with the fixture definition (this needs more testing) **This no longer seems to be true instead it seems to be controlled by the simulator Options**



Within the **Simulator Options** you can choose to "use stock setup" which will cause it to use stock as defined within the **machine group stock parameters**.

Exporting a stock model for use as stock etc in machine sim...





Begin Simulation

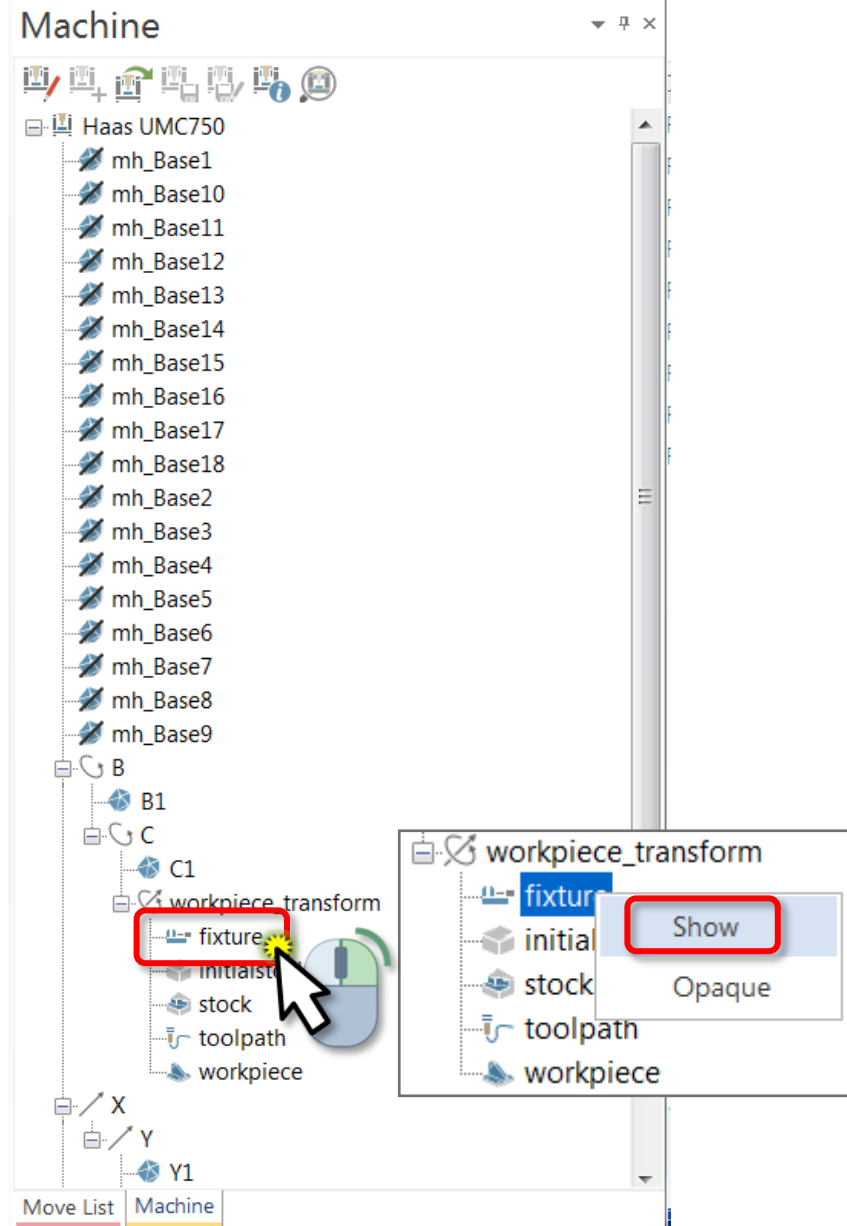


The simulate button is used to actually begin simulation, the green check mark only stores the settings and closes the dialogue box.

Modifying the Machine

Adding a Fixture element

Also note that at one time the fixture did not show, it needed to be added to the machine tree within the machine sim. Not meaning we added the specific fixtures from the part file but a generic fixture element attached to the platter. This only had to be done once(?).



Modifying the Travel

The screenshot shows the 'Machine' configuration window. The top section displays a tree view of machine components. A red circle with the number '1' highlights the 'Machine' icon at the top left. A red circle with the number '2' highlights the 'X' axis icon in the tree view, with a callout box that says 'Choose the axis you wish to modify'. The tree view includes components like 'fixture', 'initialstock', 'stock', 'toolpath', 'workpiece', 'Y1', 'Z', 'Z1', 'Z2', 'Z3', 'Z4', 'Z5', 'Z6', 'holder_transform', 'tool', 'Holder Part', 'Arbor Part', 'Shaft Part', 'Flute Part', and 'cc1'. The bottom section shows a configuration table for the selected axis (X). A red circle with the number '3' highlights the 'Min' value in the 'Limits' section, with a callout box that says 'Left click to begin modifying the value, type the desired new value.'.

Machine

1

2 Choose the axis you wish to modify

Y1

Z

Z1

Z2

Z3

Z4

Z5

Z6

holder_transform

tool

Holder Part

Arbor Part

Shaft Part

Flute Part

cc1

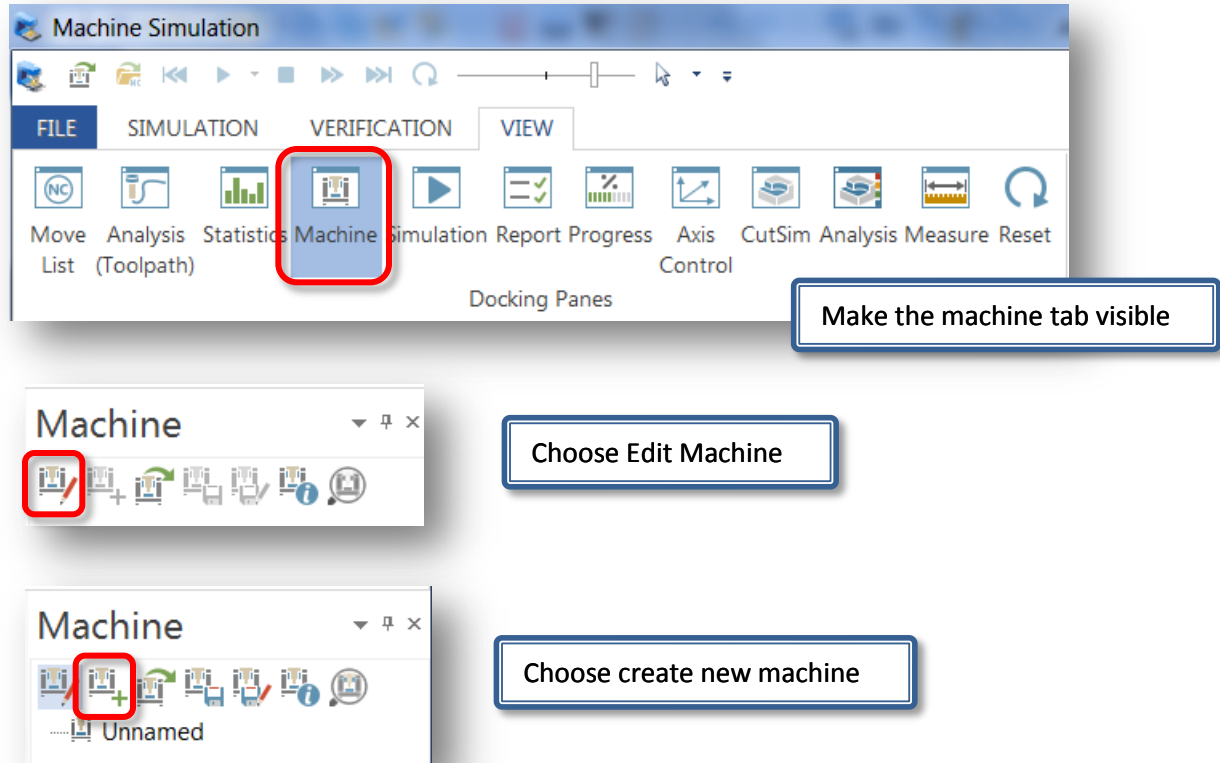
ID	X
Show slider	True
Direction	
X	1.000000
Y	0.000000
Z	0.000000
Limits	
Min	-18.987500
Max	11.012500
Values Type	Continuous
Initial Value	0.000000

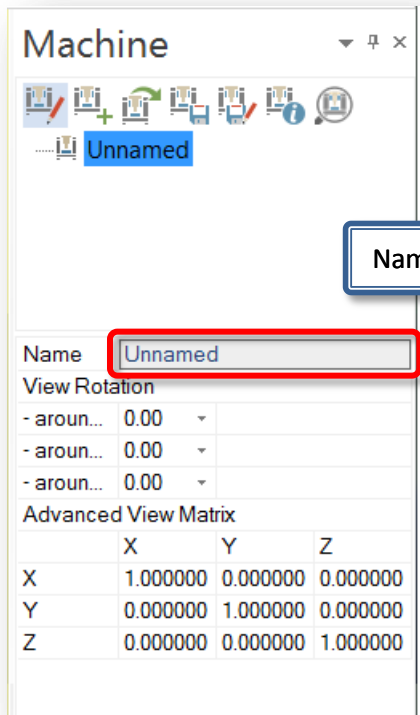
3 Left click to begin modifying the value, type the desired new value.

Move List Analysis Statistics Machine Simulation

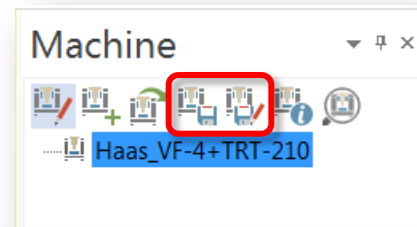
Creating a Machine From Scratch

1. It seems best to start by creating a folder for the machine within the machine simulation folder (for example C:\Users\Public\Documents\shared Mcam2018\MachineSimulation\MachSim). The folder name will need to match the name you give the machine when you save it (the XML), and it should be saved in the folder you created. This is also true of the .gif file (used as the illustration when browsing the list of available machines for mach sim). You should also put the stl files you intend to use for the machine model in that folder as well.
2. So far the only way I have found to begin creating a new machine for machine sim (also true of editing), is to first begin simulation from within mastercam in the same way you would to actually run a machine sim (see [this section](#), the setup prior to simulation isnt really relevant in this case however and can mostly be disregarded), meaning their must be toolpaths to actually output for the machine sim within the mastercam file).

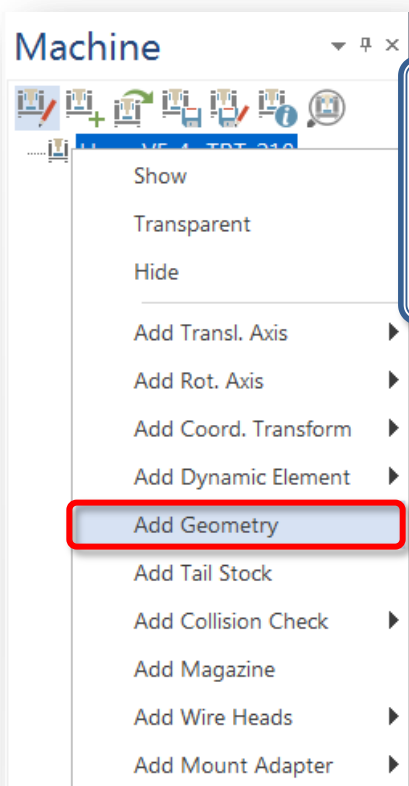




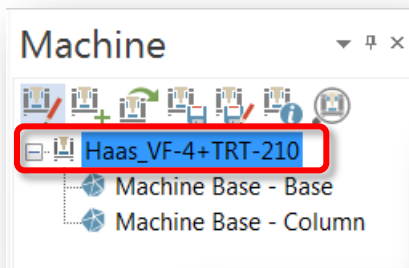
Name the Machine



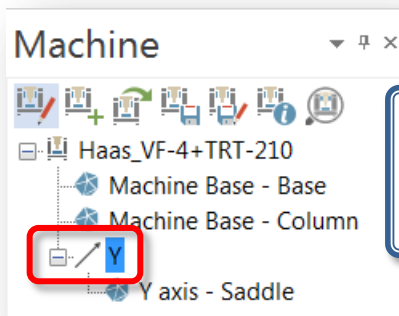
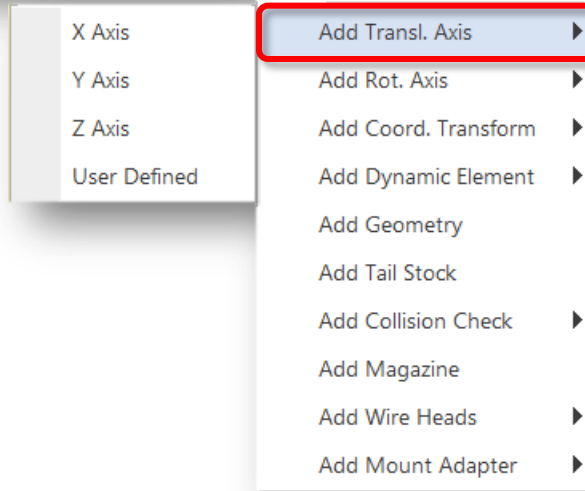
Save the machine in the folder you created for it. Note that if you didn't put the STLs for the machine in that folder as you add them to the machine they will be saved in that folder whenever the machine is saved.



Right click the machine name in the tree and choose **add geometry**. Choose the STL files to use for the machine base. You can use multiple STL files but must add the one at a time. Each STL file can contain more than one model though.

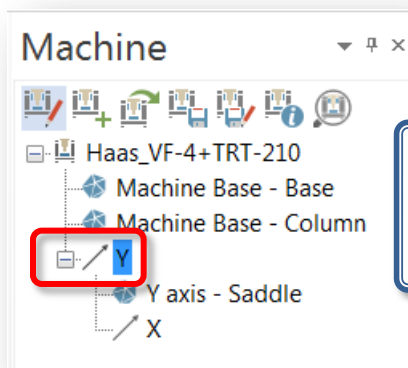


Right click the machine name in the tree and choose **add transl. Axis** to add a linear axis.

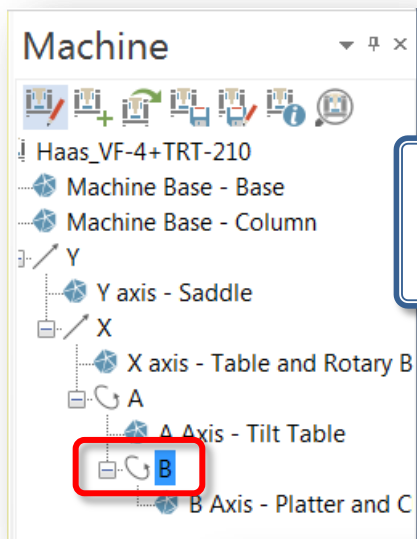


Right click the new axis and choose **add geometry** to choose an STL to represent it

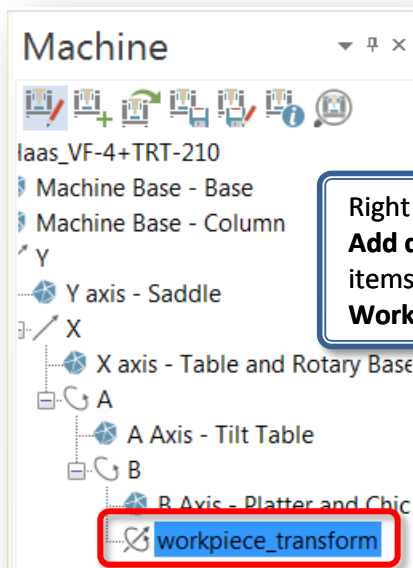
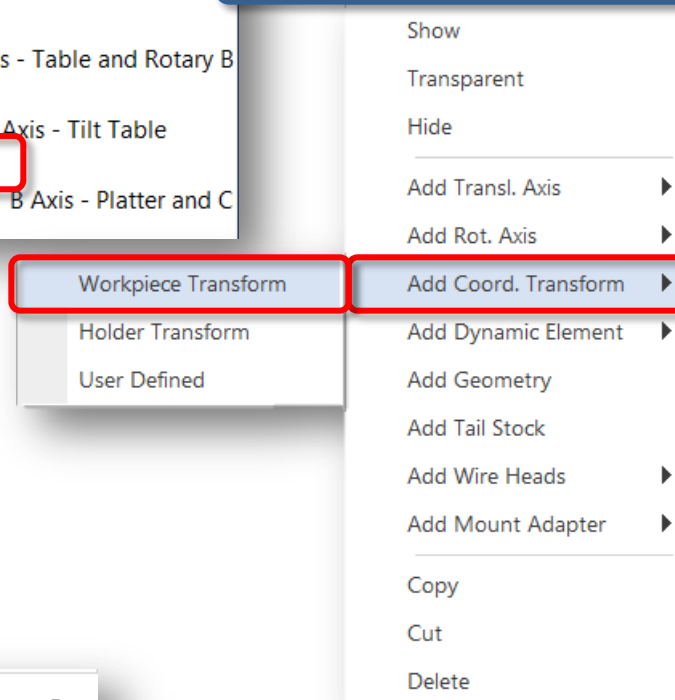
NOTE: IF YOU PLAN TO HAVE THE PLATTER OF A TRUNION PARALLEL TO THE XY PLANE WHEN IT IS AT TILT ZERO POSITION THAT IS CONSIDERED A 'C' AXIS EVEN THOUGH HAAS TRUNIONS CAN SOMETIMES CALL IT A 'B' AXIS



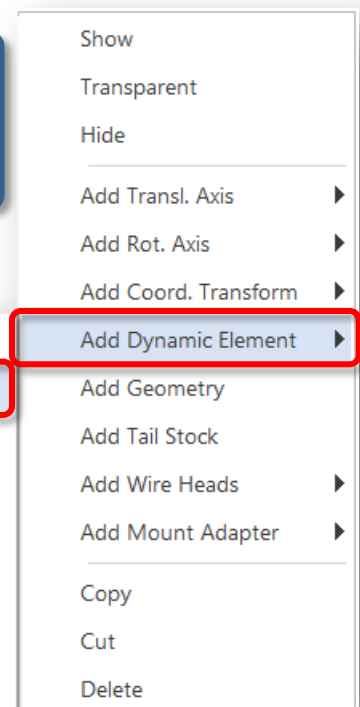
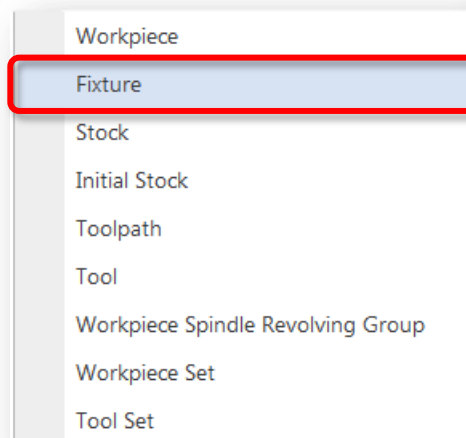
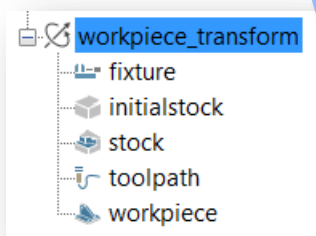
Right click the new axis and choose **add Transl. Axis** or **Add Rot. Axis** to stack another axis on it

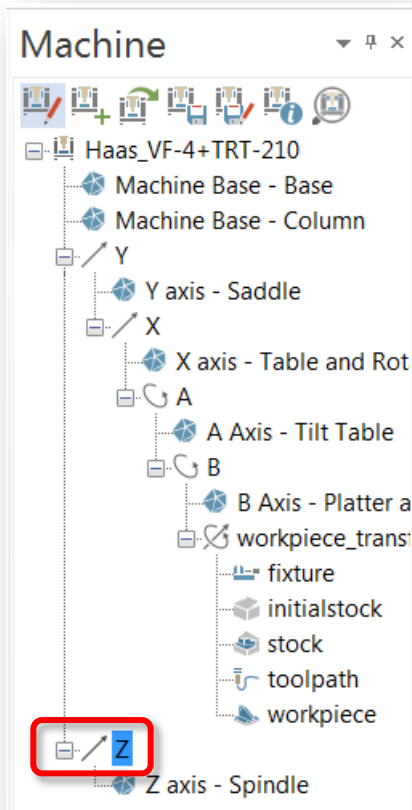


When you want to add the workpiece, Fixture etc, right click the Axis you want to add it to and choose **Add Coord. Transform** then **Workpiece Transform**

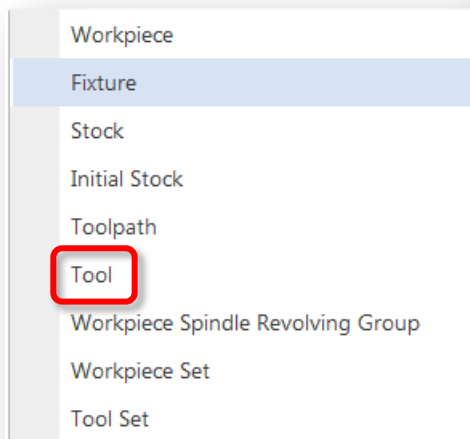
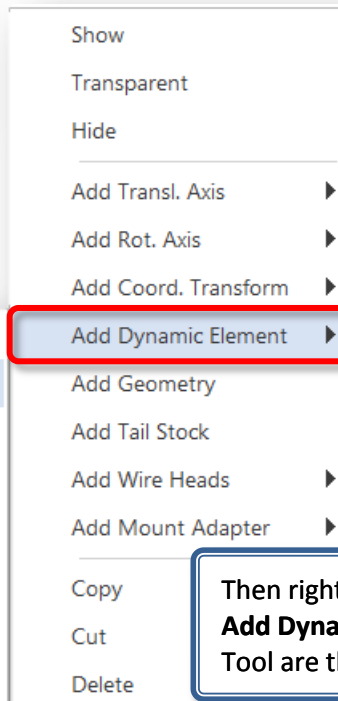
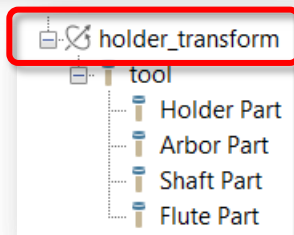
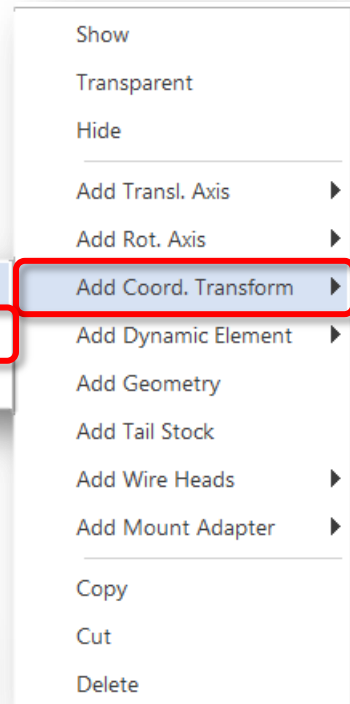
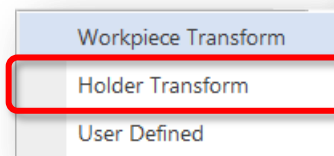


Right click the **Workpiece Transform** and choose **Add dynamic element**. Then choose each of the items shown here one by one to add to the **Workpiece Transform**.



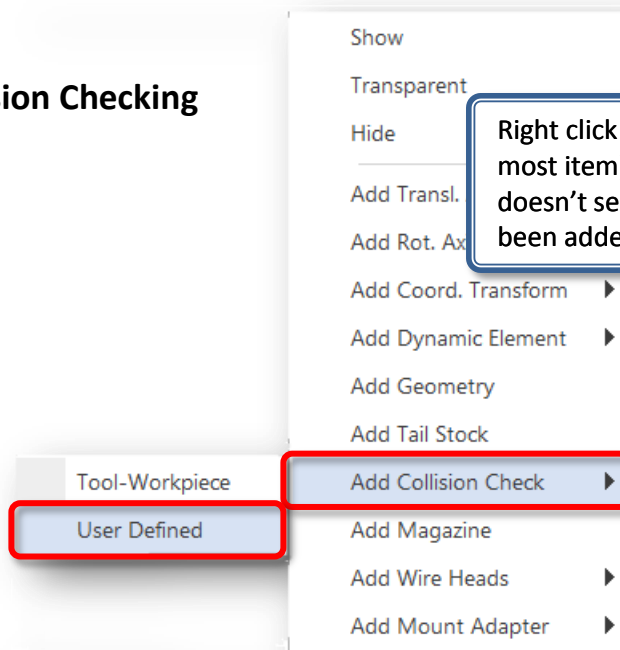


When you want to add the tool, right click the Axis to add it to and choose **Add Coord Transform** then **Holder Transform**. Transform.

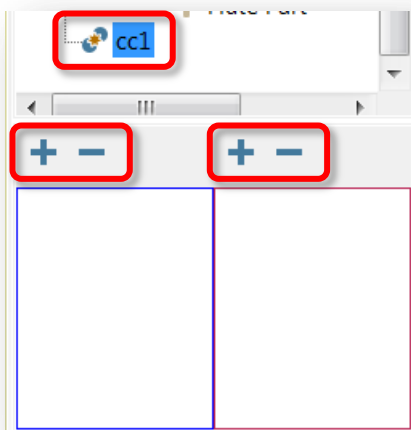


Then right click the **Holder Transform** and choose **Add Dynamic Element**, then **Tool**. The items below Tool are then added automatically as well.

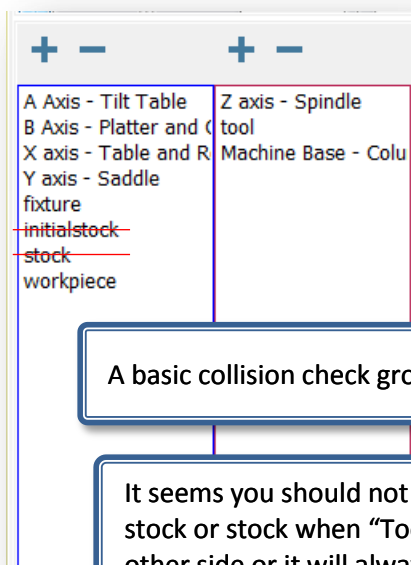
Collision Checking



Right click the entire machine in the tree (the top most item) and choose **Add Collision Check**. It doesn't seem to appear until certain elements have been added to the machine

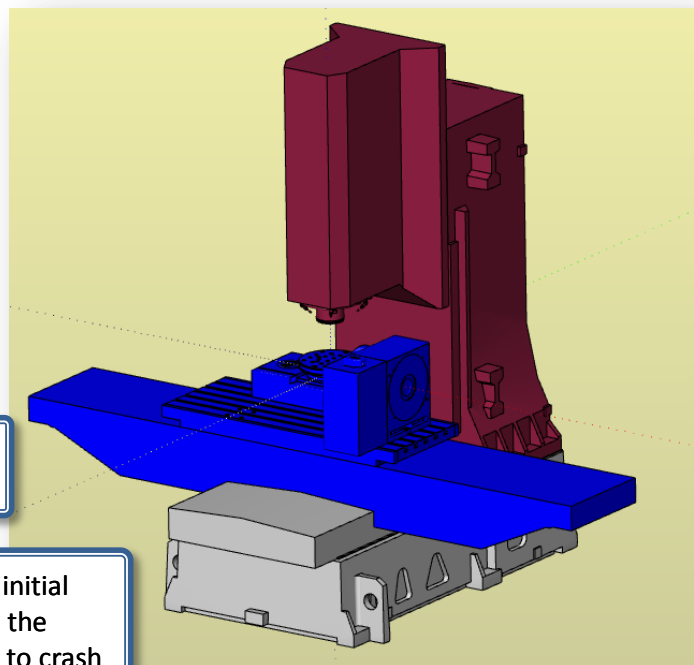


Then choose the newly added **Collision Check** to build two groups of entities to check for collision. Use the + and – signs to add or remove entities from each group



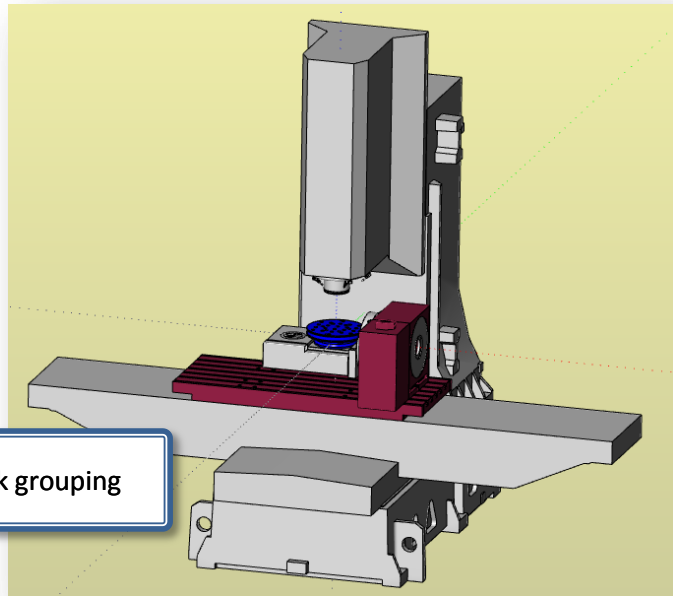
A basic collision check grouping

It seems you should not include initial stock or stock when "Tool" is on the other side or it will always seem to crash



+ - + -	
B Axis - Platter and fixture	X axis - Table and R
initialstock	
stock	
workpiece	

A secondary collision check grouping



Units (Metric vs Inch)

David from tech support was under the impression all machines had to be in metric units but he wasn't aware of the ability to change that from within the xml file of the machine. If you rename the xml as .txt you can edit it in a text editor and replace the word "metric" with "inch". Not sure if this is a bad idea or not.